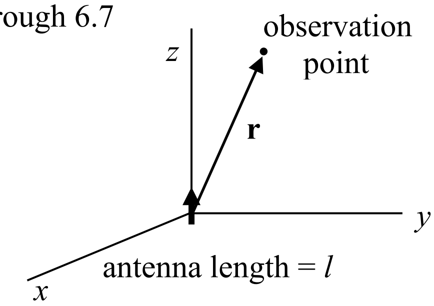
ECEN 635 **Notes #6** Sect Balanis Read 6.6 through 6.7



Far Field Approximations and Antenna Descriptives

Recall that for a infinitesimal current source

$$A_z = \frac{\mu I \ell e^{-jkr}}{4\pi r}$$

$$\mathbf{E} = -\frac{I\ell k^2}{4\pi}\eta\left\{\left[\frac{1}{(jkr)^2}+\frac{1}{(jkr)^3}\right]e^{-jkr}2\cos\theta\hat{\mathbf{r}} + \left[\frac{1}{jkr}+\frac{1}{(jkr)^2}+\frac{1}{(jkr)^3}\right]e^{-jkr}\sin\theta\hat{\boldsymbol{\theta}}\right.$$

$$\mathbf{H} = -\frac{I\ell k^2}{4\pi}\left[\frac{1}{jkr}+\frac{1}{(jkr)^2}\right]\sin\theta e^{-jkr}\hat{\boldsymbol{\varphi}}$$

Let $r \to \infty \Rightarrow$ drop all $\frac{1}{r^2}$ and $\frac{1}{r^3}$ terms

$$\mathbf{E} = j\omega\sin\theta\frac{\mu I\ell e^{-jkr}}{4\pi r}\hat{\boldsymbol{\theta}} = j\omega\sin\theta A_z\hat{\boldsymbol{\theta}}$$

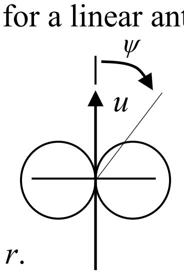
so $\quad \mathbf{E} = j\omega\sin\theta A_z\hat{\boldsymbol{\theta}} \qquad (1)$

also $\quad \mathbf{H} = \frac{j\omega\sin\theta}{\eta}A_z\hat{\boldsymbol{\varphi}} \qquad (2)$

$\Rightarrow \quad H_\phi = \frac{E_\theta}{\eta} \qquad (3)$

These "far field" approximations (eqn.1, 2, and 3) work for a linear antenna of any length (except infinite) that lies along the $z-$axis.

In general for an antenna lying on the u-axis
$\mathbf{E} = j\omega\sin\psi A_u\hat{\boldsymbol{\psi}}$

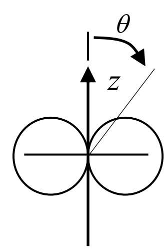


Radiation intensity pattern $\equiv |E_\theta|$ at a constant distance r.

$$|E_\theta| = \left|\frac{E_\theta}{E_\theta(\max\theta,\phi)}\right|$$

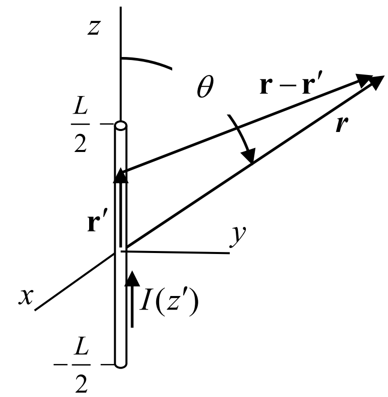
Example: short dipole

$$|E_\theta| = \left|\frac{j\omega\mu\sin\theta\frac{I\ell e^{-jkr}}{4\pi r}}{j\omega\mu\frac{I\ell e^{-jkr}}{4\pi r}}\right| = |\sin\theta|$$



The general expression for the potential of a linear current element is:

$$A_z = \mu \int_{vol} J(z') \frac{e^{-jk|\mathbf{r}-\mathbf{r}'|}}{4\pi|\mathbf{r}-\mathbf{r}'|} dv'$$



Approximations for $|\mathbf{r}-\mathbf{r}'|$

$$\begin{aligned}|\mathbf{r}-\mathbf{r}'| &= \left[(\mathbf{r}-\mathbf{r}')\cdot(\mathbf{r}-\mathbf{r}')\right]^{1/2} \\ &= \left[r^2 - 2\mathbf{r}\cdot\mathbf{r}' + r'^2\right]^{1/2} \\ &= r\left[1 - 2\frac{\mathbf{r}\cdot\mathbf{r}'}{r^2} + \left(\frac{r'}{r}\right)^2\right]^{1/2}\end{aligned}$$

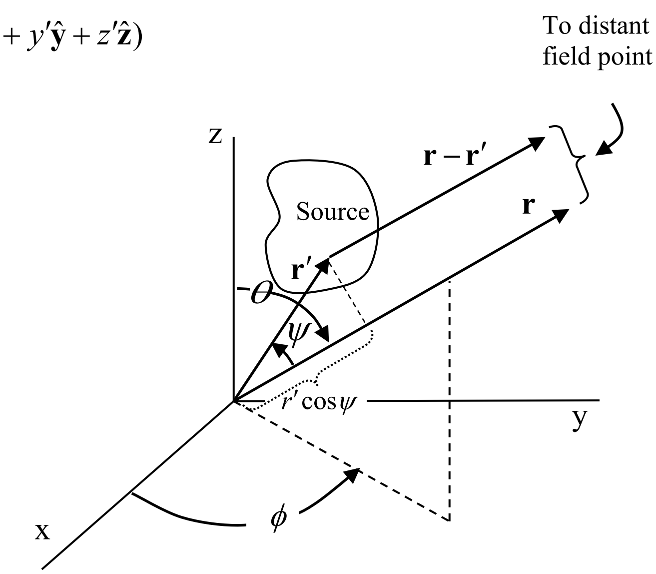
Use $\sqrt{1+x} = 1 + \frac{x}{2} - \frac{x^2}{8} + \cdots$

$$\begin{aligned}|\mathbf{r}-\mathbf{r}'| &= r\left[1 + \frac{1}{2}\left(-2\frac{\mathbf{r}\cdot\mathbf{r}'}{r^2} + \left(\frac{r'}{r}\right)^2\right) - \frac{1}{8}\left(-2\frac{\mathbf{r}\cdot\mathbf{r}'}{r^2} + \left(\frac{r'}{r}\right)^2\right)^2 + \ldots\right] \\ &= r - \frac{\mathbf{r}\cdot\mathbf{r}'}{r} + \text{(higher order terms)} \\ &\cong r - \hat{\mathbf{r}}\cdot\mathbf{r}'\end{aligned}$$

Far field approximation for $|\mathbf{r}-\mathbf{r}'| \to r - r'\cos\psi$ as presented in Balanis.

Source region in **Cartesian coordinates**, observation point in spherical coordinates:

$$\begin{aligned}|\mathbf{r}-\mathbf{r}'| &\cong r - \hat{\mathbf{r}}\cdot\mathbf{r}' = r - r'\cos\psi \\ &= r - \left(\sin\theta\cos\phi\,\hat{\mathbf{x}} + \sin\theta\sin\phi\,\hat{\mathbf{y}} + \cos\theta\,\hat{\mathbf{z}}\right)\cdot(x'\hat{\mathbf{x}} + y'\hat{\mathbf{y}} + z'\hat{\mathbf{z}}) \\ &= r - x'\sin\theta\cos\phi - y'\sin\theta\sin\phi - z'\cos\theta\end{aligned}$$



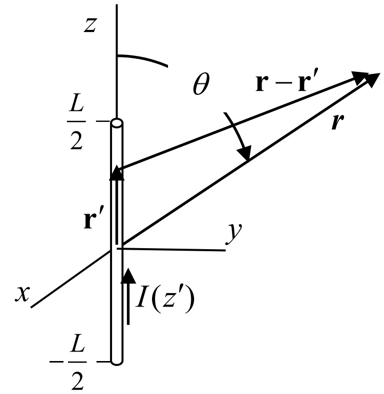
Source region in **cylindrical coordinates**, observation point in spherical coordinates:

$$\begin{aligned}|\mathbf{r}-\mathbf{r}'| &\cong r - \hat{\mathbf{r}}\cdot\mathbf{r}' = r - r'\cos\psi \\ &= r - \rho'\sin\theta\cos(\phi-\phi') - z'\cos\theta\end{aligned}$$

Source region in **spherical coordinates**, observation point in spherical coordinates:

$$\begin{aligned}|\mathbf{r}-\mathbf{r}'| &\cong r - \hat{\mathbf{r}}\cdot\mathbf{r}' = r - r'\cos\psi \\ &= r - r'\left[\cos\theta\cos\theta' + \sin\theta\sin\theta'\cos(\phi-\phi')\right] \qquad (3-102)\end{aligned}$$

Field of a long linear antenna.



For an infinitesimal source we can approximate

$$\left|\mathbf{r} - \mathbf{r}'\right| \doteq r \qquad (4)$$

because $r \gg r'$ whose maximum value is $l/2$ with $l \ll \lambda$

But for a long linear antenna $L > \lambda/10$, significant phase error can result if $e^{-jk|\mathbf{r}-\mathbf{r}'|} \cong e^{-jkr}$ is used. So from the previous page use the phase

$$\left|\mathbf{r} - \mathbf{r}'\right| \cong r - z'\cos\theta \qquad (5)$$

$$\therefore A_z = \mu \int_{vol} J\left(z'\right)\frac{e^{-jk|\mathbf{r}-\mathbf{r}'|}}{4\pi\left|\mathbf{r}-\mathbf{r}'\right|}dv' \cong \frac{\mu e^{-jkr}}{4\pi r}\int_{-L/2}^{L/2} I\left(z'\right)e^{jkz'\cos\theta}dz' \qquad (6)$$

The current approximation $I\left(z\right) = I_o \sin\left[k\left(\frac{L}{2} - |z|\right)\right]$ in (6) gives

$$A_z = \frac{\mu I_o e^{-jkr}}{4\pi r}\left\{\int_{-L/2}^{0} \sin\left[k\left(\frac{L}{2}+z'\right)\right]e^{jkz'\cos\theta}dz' + \int_{0}^{L/2} \sin\left[k\left(\frac{L}{2}-z'\right)\right]e^{jkz'\cos\theta}dz'\right\}$$

$$= \frac{\mu I_o e^{-jkr}}{2\pi r}\frac{\left[\cos\left(k\,L/2\cos\theta\right) - \cos\left(k\,L/2\right)\right]}{k\sin^2\theta} \qquad (7)$$

The electric field

$$E_\theta = j\omega\sin\theta A_z$$

so

$$E_\theta = \frac{j\eta I_o}{2\pi}\frac{e^{-jkr}}{r}\left[\frac{\cos\left(k\,L/2\cos\theta\right) - \cos\left(k\,L/2\right)}{\sin\theta}\right] \qquad (8)$$

Radiated power density

$$\mathbf{S} = \frac{1}{2}\mathbf{E}\times\mathbf{H}^{*} \Rightarrow S_r = \frac{1}{2}E_\theta H_\phi^{*} = \frac{E_\theta E_\theta^{*}}{2\eta} \qquad (9)$$

$$S_r = \frac{\eta I_o^2}{2(2\pi r)^2}\left[\frac{\cos(k\,L/2\cos\theta)-\cos(k\,L/2)}{\sin\theta}\right]^2 \text{ for a long linear antenna}$$

The Hertzian dipole radiated power density:

$$S_r = \frac{E_\theta E_\theta^{*}}{2\eta} = \frac{\eta}{2}\left(\frac{I_o L k}{4\pi}\right)^2 \frac{\sin^2\theta}{r^2} \quad \text{where } E_\theta = j\omega\sin\theta\frac{\mu I_o L e^{-jkr}}{4\pi r}$$

Total radiated power

$$P_f = \int_{\substack{\text{spherical}\\ \text{surface}}} \mathbf{S}\cdot d\mathbf{A} = \int_{\phi=0}^{2\pi}\int_{\theta=0}^{\pi} S_r\; r^2\sin\theta\, d\theta\, d\phi \qquad (10)$$

$$= \eta\frac{I_o^2}{4\pi}\int_0^{\pi}\frac{\left[\cos(k\,L/2\cos\theta)-\cos(k\,L/2)\right]^2}{\sin^2\theta}d\theta \text{ for a long linear antenna}$$

and for a Hertzian dipole

$$P_f = \frac{\pi\eta}{3}I_o^2\left(\frac{L}{\lambda}\right)^2 \qquad \text{where } L \ll \lambda/2$$

Radiation resistance

$$R_r = \frac{P_f}{|I_\mathrm{m}|^2}\ , \quad \text{where } I_\mathrm{m} \text{ is the current maximum on the antenna} \qquad (11)$$

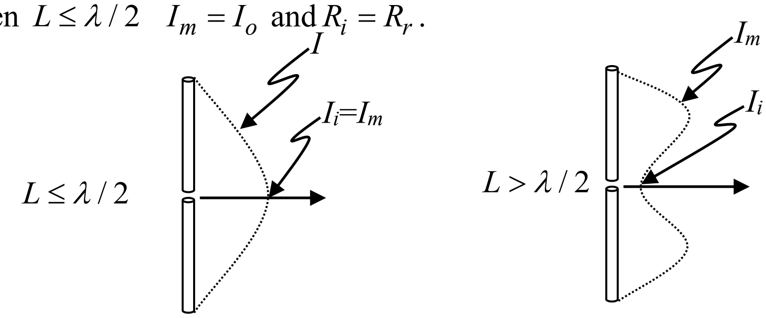
$$= \frac{\eta I_o^2}{I_m^2 4\pi}\int_0^{\pi}\frac{\left[\cos(k\,L/2\cos\theta)-\cos(k\,L/2)\right]^2}{\sin^2\theta}d\theta \text{ for a long linear antenna}$$

Input resistance

$$R_i = \frac{P_f}{|I_i|^2}, \quad \text{where } I_i \text{ is the current amplitude at the antenna feed point} \qquad (12)$$

so in general $R_i = \dfrac{|I_m|^2}{|I_i|^2}R_r$ for a long linear dipole antenna and

when $L \le \lambda/2 \quad I_m = I_o$ and $R_i = R_r$.

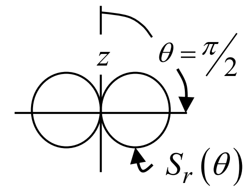


Directivity

$$D_o = \frac{\text{max time ave power density}}{\text{ave power though a sphere}} = \frac{S_r(\theta,\varphi)_{\max}}{\dfrac{P_f}{4\pi r^2}} = \frac{4\pi r^2 S_r(\theta,\varphi)_{\max}}{P_f} \qquad (13)$$

For a Hertzian dipole:

$$D_o = \frac{4\pi r^2 \left[\eta \left(\dfrac{I_o lk}{4\pi}\right)^2 \dfrac{\sin^2\theta}{2r^2}\right]_{\theta=90^o}}{\dfrac{\eta\pi}{3} I_o^2 \left(\dfrac{l}{\lambda}\right)^2} = 1.5$$



Efficiency

Radiation efficiency: $e_r = \dfrac{P_f}{P_{in}} = \dfrac{P_f}{P_f + P_\Omega} = \dfrac{I_o^2 R_r/2}{I_o^2 R_r/2 + I_o^2 R_\Omega/2} = \dfrac{R_r}{R_r + R_\Omega}$ (14)

An antenna with no ohmic losses $R_\Omega = 0$, and $e_r = \dfrac{R_r}{R_r + 0} = 1$

A dipole antenna with ohmic losses $R_\Omega = \dfrac{\rho L}{A} = \dfrac{L}{\sigma 2\pi a \delta_d}$

conductivity of the antenna metal
antenna wire radius
skin depth

Reflection efficiency $e_{ref} = (1 - |\Gamma|^2)$ where $\Gamma = \dfrac{Z_L - Z_o}{Z_L + Z_o}$ (15)

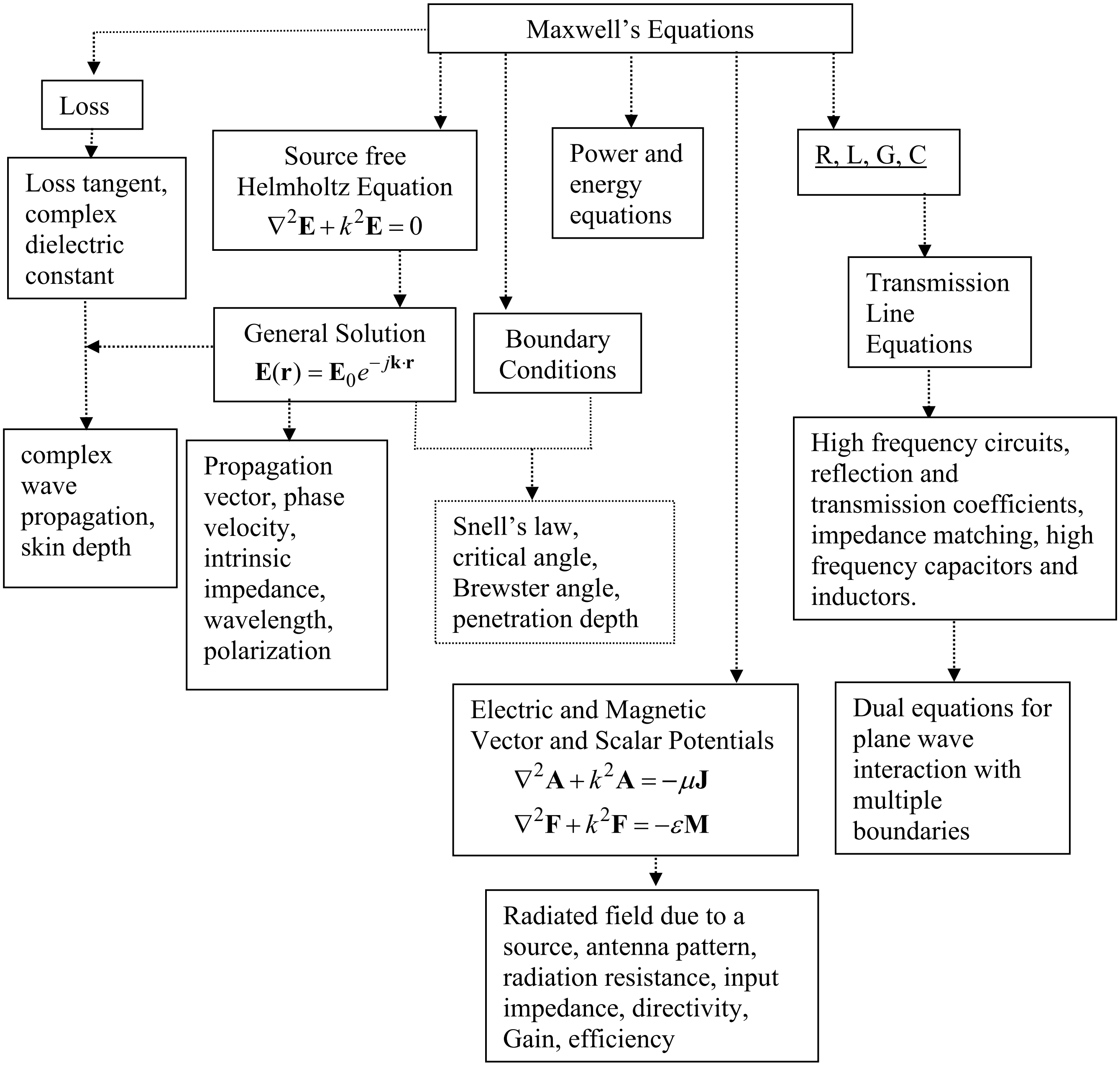
Ideally, at resonance the antenna is impedance matched to the feed line $Z_L = Z_o$ so $e_{ref} = 1$

Antennas can have other efficiencies that you learn about when you take an antennas course.

Gain $\equiv$ Directivity $\times$ efficiencies

$$G = D_o e_{ref} e_r \qquad (16)$$

If both $e_r = 1$ and $e_{ref} = 1$ then $G = D_o$.


Maxwell's Equations
Loss
Loss tangent, complex dielectric constant
Source free Helmholtz Equation
$\nabla^2\mathbf{E}+k^2\mathbf{E}=0$
Power and energy equations
R, L, G, C
Transmission Line Equations
General Solution
$\mathbf{E}(\mathbf{r})=\mathbf{E}_0 e^{-j\mathbf{k}\cdot\mathbf{r}}$
Boundary Conditions
complex wave propagation, skin depth
Propagation vector, phase velocity, intrinsic impedance, wavelength, polarization
Snell's law, critical angle, Brewster angle, penetration depth
High frequency circuits, reflection and transmission coefficients, impedance matching, high frequency capacitors and inductors.
Electric and Magnetic Vector and Scalar Potentials
$\nabla^2\mathbf{A}+k^2\mathbf{A}=-\mu\mathbf{J}$
$\nabla^2\mathbf{F}+k^2\mathbf{F}=-\varepsilon\mathbf{M}$
Dual equations for plane wave interaction with multiple boundaries
Radiated field due to a source, antenna pattern, radiation resistance, input impedance, directivity, Gain, efficiency